1. (10 points) Special Spanning Tree

Given a weighted undirected graph $G = (V, E)$ and an integer k. Other than normal graphs, some edges are defined as "special" in the graph. Each edge in this graph can be represented as a quadruple: $e_i = (u_i, v_i, w_i, s_i)$. u_i and v_i mean the indices of the vertices connected by the edge, w_i means the weight of the edge, and s_i indicates whether the edge is a special edge(i.e. $s_i = 1$ means it's special and $s_i = 0$ means it's not special).

For convenience, we agree on some denotations: E_s means the set of special edges, E_n means the set of non-special edges and $|V| = n, |E| = m, |E_s| = m_1, |E_n| = m_2$.

We want to find a spanning tree with certain requirements (Assume there exists at least one):

1. It contains exactly k special edges in the graph.
2. Among all the spanning trees satisfying the first requirement, it has minimal weight.

(In the parts below, when it comes to existing algorithms, you can describe them briefly instead of writing them down completely.)

(a) (1') Hint Question: Based on which algorithm you can solve this easily? (Notice that this spanning tree focuses more on edges rather than vertices) ◯ Prim ◯ Kruskal

(b) We start from the naive part: Consider what will happen when $k = 0$ and $k = n - 1$.

i. (2') About $k = 0$ i.e. there are no special edges in the spanning tree. Design an algorithm to solve this problem.

Solution:

ii. (2') About $k = n - 1$ i.e. there are all special edges in the spanning tree. Design an algorithm to solve this problem.

Solution:

(c) Now consider normal cases, let's find some properties by analyzing the cases above:

i. (1') In the questions above, if we increase/decrease the weights of all the elements in one set(E_n or E_s), it will not affect which edges to choose, i.e. the final spanning tree will not change. ◯ True ◯ False

ii. (1') Consider that if we only change the weights of all the elements in one set and make its relative difference unchanged (i.e. increase or decrease all values by one same value), the answer spanning tree will not change. ◯ True ◯ False

iii. (1') Assume we have a problem $P(n)$ and an answer ans we want to reach, i.e. find $n, P(n) = ans$. If $P(n)$ is monotonic with regard to n and $n \in [l, r]$, then we can use divide and conquer to solve this problem in $\Theta(\log(r - l))$. ◯ True ◯ False

(d) (2') Then we find that we can balance the number of special edges by modifying the relative difference between the two sets E_n and E_s. With the properties above, please give an algorithm to solve this problem and give out the time complexity of the algorithm. (Hint: Notice that if we increase more on weights of E_s, the number of special edges in MST will be fewer.)

Solution: